

VIP Very Important Paper



Porous Organic Compounds – Small Pores on the Rise

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In recognition of all frontline healthcare workers battling the COVID-19 pandemic

Porous materials are important for various energy-related technologies such as catalysis and separation. While porous organic cage compounds are a rather recent addition to the field of porous materials, these discrete nanocavities have since emerged as a versatile functional-materials platform, facilitated by the solubility of the materials in common organic solvents. In contrast to other frameworks, organic cages are assembled

first from modular building blocks in solution and then packed in the solid-state in a next step. In this minireview, we highlight examples of porous organic cages with focus on how the intrinsic nanopore can be controlled and utilized, especially focusing on their synthesis and the gas sorption properties of smaller intrinsic pores of porous organic cage compounds, porous macrocycles, and other related compounds.

1. Introduction

The importance of storing gases with a high global warming potential or those that are relevant in the context of renewable energies is steadily increasing, as climate change is a result of global warming leading to ecosystem damage concomitant with a decline in biodiversity worldwide.^[1] Porous materials are an important part of emerging technologies, especially for energy-related applications. The separation and purification of gases, as well as storage and release, is of great importance not only in industry but also in transportation.^[2] Novel porous materials that might be able to selectively encapsulate and release gases have therefore become the center of research efforts. Porous frameworks, bearing interconnected voids, have generated a lot of attention in the last three decades, such as metal-organic frameworks (MOFs),^[3] covalent organic frameworks (COFs),^[4] complemented by porous organic polymers (POPs) and porous organic cages (POCs).^[5] The latter are characterized particularly by their modular synthesis and their solubility compared to MOFs and COFs. This does not only significantly simplify the analysis of POCs but also opens up opportunities in regard to processing POCs for thin-film or membrane applications. Depending on the building blocks selected, surface areas similar to MOFs or COFs can be obtained with POCs, as some selected examples show.^[6] The development of new methods to analyze and determine molecular structures parallels the ability to accelerate synthetic research.^[7] Therefore, continuously improving analytical methods provides

researchers with the possibility to create and analyze larger and larger structures.^[7,8] In this minireview, we want to explore another, sometimes overlooked aspect of porous compounds – the utilization and control of small pores. The smaller a pore is, and the more functional groups are exposed within the nanopore, the stronger the interaction with a potential guest can be. In some examples presented herein, even size exclusion plays a role. Moreover, we discuss pores becoming smaller and smaller until one guest is rejected, whereas other guests can interact and pass a pore, enabling application of porous compounds for separation purposes.

Since it is not possible to cover all previously mentioned, wide areas of research in one short review we selected publications, which we deemed interesting to the general reader and according to our perception, highly relevant and timely in respect to our focus. We will concentrate mostly on smaller, self-assembled and shape-persistent porous structures, which are metal-free. We defined systems as having small pores, when the SA_{BET} is less than $1000 \text{ m}^2 \text{ g}^{-1}$ (surface area (SA), according to Brunauer-Emmett-Teller (BET) theory), since this parameter offers the best comparability across different categories of porous materials.^[9] Several excellent reviews on porous materials in general,^[9] MOFs,^[3] COFs,^[4] POCs^[5] and fluorinated organic porous materials^[10] are available for the interested reader, which offer interesting and detailed insights into the respective fields.

Although mesoporous frameworks with a large surface area are needed for potential storage and energy applications, only microporous frameworks seem to satisfy the needs of certain fields like separation technologies, catalysis and energy storage.^[11b-d] The preparation of microporous frameworks remains challenging since the crystallization of microporous 3D frameworks is difficult and 2D-frameworks often tend to form mesoporous structures.^[11a] Only a few microporous frameworks are reported and among those reported, mostly interpenetrated frameworks are found that require precise control of the synthetic conditions.^[11] To achieve microporosity in a material, the use of POCs or supramolecular organic frameworks (SOFs) might be beneficial due to their ease of synthesis and facile tuneability of the crystal lattice. Compared to other porous

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frameworks, POCs have distinct pores that ideally form accessible channels through intermolecular interactions (packing interactions) into a regular crystal lattice. Only when the windows of neighboring cage molecules are facing each other, does the (semi)crystalline material become porous.

In this regard, the dimensionality of the pore network is strongly influenced by the crystal packing and the crystallization of chemically similar molecules can result in either 0D-, 1D-, 2D- or 3D-pore networks, depending on the interactions inside the crystalline network (Figure 1).^[9c] These interactions are also the cause of the collapse of porous structures, a phenomenon that is encountered when attempting to remove the (sometimes stabilizing) solvent molecules from the cavities, in favor for attractive contacts to neighboring compounds within the network. To prevent this, compounds considered for the creation of porous networks should be rigid and comprised of stable bonds. Only cage compounds that remain unchanged in their morphology during activation, gas adsorption, and desorption can be deemed “shape-persistent” and are therefore suitable for a porous material.^[5f] The resulting porous network can either consist of the intrinsic pore, that is the inner cavity of each POC, or of the extrinsic pore, which is a result of inefficient packing due to e.g. steric hindrance within the crystal lattice. It is important to keep in mind, that data obtained from single-crystals does not necessarily represent the bulk material, therefore the morphology and stability of these networks is ideally monitored by powder X-ray diffraction (PXRD) before and after thermal activation. During gas sorption measurements, a non-linear DFT-derived (NL-DFT) pore size distribution can be obtained to indicate whether the most abundant pore, responsible for porosity, is the extrinsic or intrinsic pore, since defects within the lattice can also contribute to porosity.^[9d] Although size exclusion phenomena can be achieved by thorough engineering of the extrinsic pore, the more rational approach would be to gain control of the intrinsic pore size and properties, ideally without influencing the extrinsic pore.^[9] As

the majority of organic cages presented herein are formed by reversible dynamic covalent chemistry (DCC), the resulting cages will be labelled according to the nomenclature proposed by Jelfs and co-workers.^[12] The $X^m Y^n$ convention is based on the type, number, and multiplicity of connectivity of each structure's building blocks, where X and Y are the two different building blocks that comprise the cage. If X and Y are ditopic, tritopic, or tetratopic, they are labeled Di, Tri, and Tet, respectively. For X and Y, the superscripts m and n denote the number of each build incorporated into the topology.

2. Porous Organic Cages

To be able to influence the interactions of gases or guests within the intrinsic (inside) pore, steric hindrance can be used to exclude molecules of a particular size. Therefore, it is possible to either change i) the window size or ii) the size of the nanocavity itself. Cooper *et al.* synthesized POCs not only from condensation of amines with aldehydes, but also from amines with ketones (Figure 2).^[13] When reacting (1*R*,2*R*)-(–)-1,2-diaminocyclohexane (**R-1**) with 1,3,5-triacetylformylbenzene (**2**), they were able to synthesize a Tri^4Di^6 -cage with methyl groups attached to each of the imine groups, leading to a reduced window size. Since the Me- Tri^4Di^6 (**CC15**) cage did not crystallize in a favorable window-to-window packing fashion, Cooper *et al.* created cocrystals of **CC3** and **CC15** with a *quasiracemic* composition (Figure 2). In the cocrystal, **CC15** and **CC3** were occupying alternating positions in the crystal lattice with a window-to-window packing. The reduced window size of **CC15** lead to selective porosity for H_2 at 77 K within the cocrystal, whereas for N_2 no noticeable porosity could be measured. Furthermore, the group introduced cage **CC19**, in which the triformylbenzene building blocks (**3**) bear one additional hydroxyl group at the core (**4**). Again, this led to a reduced window size compared to **CC3**, although the window was much



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Bernd M. Schmidt researches responsive and functional supramolecular systems. Born and raised in Berlin, Germany, he did his PhD with Dieter Lentz at the Freie Universität Berlin and with Hidehiro Sakurai at the IMS in Okazaki, Japan, working on bowl-shaped aromatic hydrocarbons. Bernd stayed with a Humboldt research fellowship with Makoto Fujita at the University of Tokyo, Japan and with Stefan Hecht at the HU Berlin, Germany. Since February 2018, he has been an independent research group leader at the HHU Düsseldorf in Germany, where he became a Member of the Young Academy of the North Rhine-Westphalian Academy of Sciences, Humanities and the Arts in 2020.

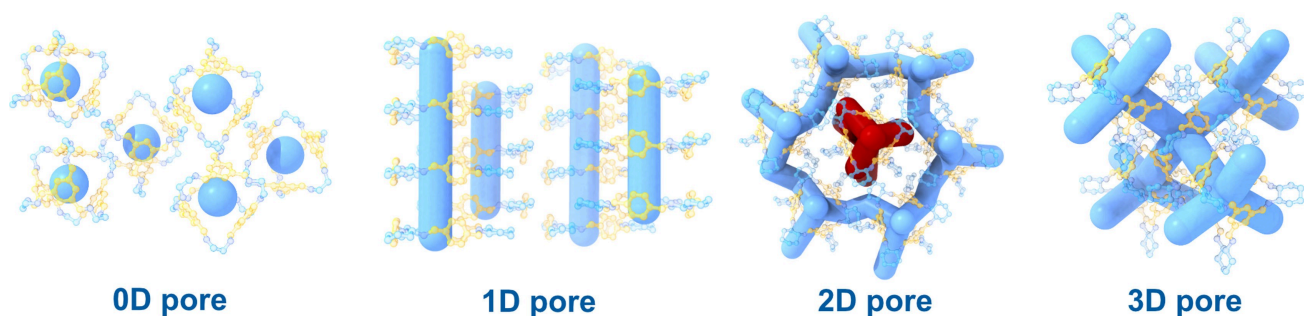


Figure 1. Schematic representation of different types of intrinsic pores in organic compounds can be used to classify the porosity in organic crystals; isolated 0D pores, channel-like 1D pores, interconnected 2D and a highly interconnected 3D pore structures (from left to right).^[9c]

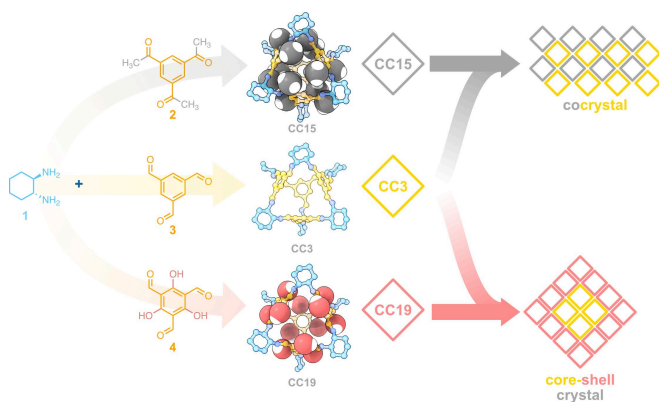


Figure 2. Cocystal formation in the solid-state blending shape-persistent Tri⁴Di⁶ POC CC15 and CC3, assembled using 1,3,5-triacetylbenzene (**2**) and triformylbenzene (**3**), respectively (top);^[13] synthesis of POC CC19 with 2-hydroxy-1,3,5-benzenetricarbaldehyde (**4**), with disordered hydroxyl groups occupying the four cage windows (red space-filling model, bottom).^[13] Hydrogen atoms and solvent molecules are omitted for clarity, space-filling view including hydrogen atoms for *endo*-groups.

larger than that of CC15. When combining CC19 and CC3 in a core-shell material, the researchers were able to selectively block the pores of the core cage CC3 for CH₄ entry by size exclusion, using the smaller pore windows of CC19 within the shell (Figure 2).

Controlling the window size by introducing residues either at the core of the triformyl building block or substituting the aldehydes with methyl ketones was employed to control gas selectivities in porous organic liquids or to separate chiral compounds using cage-coated columns in gas chromatography applications.^[13b,c]

To address the principal need for a POC with a desired pore size and favorable electronic properties for the specific binding of guests, the modification of functional groups within already known cages is obviously preferable over a *de novo* synthesis. The latter approach needs to overcome the often-encountered problem of laborious systematic optimization that is required when exploring the material space of a previously unknown POC.^[12a]

Such modification can either happen at the periphery of the imine groups, which increases stability and crystallinity, but is

expected to only yield a small influence on the binding behavior inside the cavity, or employing functional groups that point towards the center of the cavity, like amino or hydroxyl groups.^[14,15] Mastalerz *et al.* were among the first to selectively functionalize reactive hydroxyl groups inside the cavity after successful synthesis of a salicylbisimine Tri⁴Di⁶ POC SC1 (Figure 3). When reacting the exposed hydroxyl groups by means of a sixfold Williamson ether synthesis, alkylation of all six hydroxyl groups proceeded in very good yields with ether moieties ranging from simple methyl groups to 4-nitrobenzyl groups.

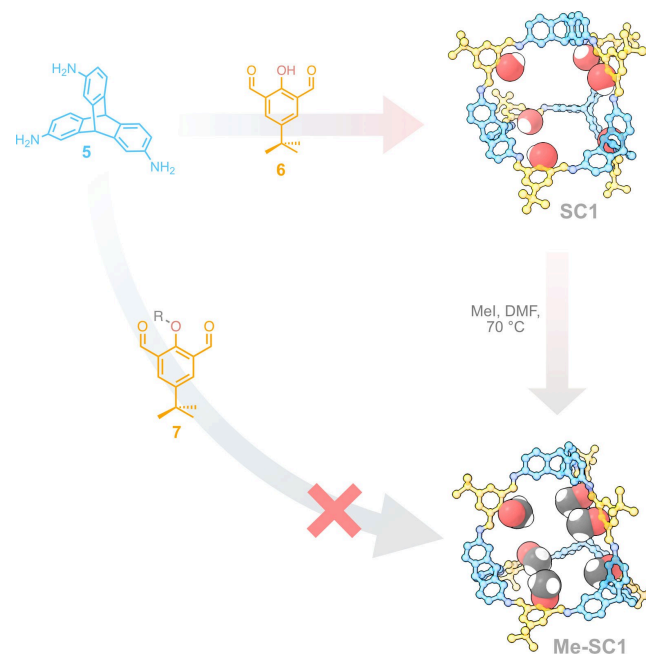


Figure 3. Salicylbisimine Tri⁴Di⁶ POC by the group of Mastalerz, showing the exposed hydroxyl groups within the nanocavity (red space-filling model). While post-synthetic methylation within the cage's cavity is possible in solution (red and grey space-filling model, assembly yield 58%,^[5f] methylation 81% yield, overall two-step yield 47%), direct assembly of the methylated building block (**7** R=Me) resulted only in a small amount of cage Me-SC1 in the crude product, never exceeding 17% yield even after optimisation attempts.^[14d] Hydrogen atoms and solvent molecules are omitted for clarity, space-filling view including hydrogen atoms for *endo*-groups.

The methyl-functionalized POC **Me-SC1** showed a slightly lower BET area compared to the parent OH-derivative **SC1**, but exhibited a higher selectivity towards CH_4 over CO_2 . This behavior can be rationalized by the less polar methoxy groups, which showed a higher affinity to hydrophobic gases like CH_4 ,^[14d] making this an early example of very successful cavity functionalization of POCs (*vide infra*).

When Cooper and co-workers intended to crystallize the reduced derivative **RCC3** of their Tri^4Di^6 imine POC (**CC3**) from an acetone solution, they were surprised to find a non-collapsed, so-called “acetone-tied” structure (**AT-RCC3**),^[14c] resulting from a reaction published by Gawronski and Rychleswka.^[15] Only one acetone molecule per cage reacts with two adjacent amine groups to form an aminal, significantly increasing the stability, crystallinity and porosity of the cage. Although **AT-RCC3** comprised a non-collapsed cavity, the porosity towards N_2 and H_2 was significantly decreased, as the methyl groups blocked approximately half the pore’s windows (Figure 4).

However, the affinity towards CO_2 just below room temperature, increased eightfold over the parent cage **CC3**. Thermal motion seemed to be the driving force for gas molecules to overcome blocked pathways inside the pore’s channels of the **AT-RCC3** material. As an answer towards the unintentional blocking of the pore’s windows, the researchers succeeded in preparing **6FT-RCC3**, a “formaldehyde-tied” congener of **CC3**, in which all amines were reacted with formaldehyde. The aminal

POC reprises the geometry of **CC3** almost identically and crystallizes in a similar lattice. The specific surface area of **6FT-RCC3** also rivals that of the parent imine cage, ($S_{\text{BET}} = 409 \text{ m}^2 \text{ g}^{-1}$ for **CC3**, $S_{\text{BET}} = 377 \text{ m}^2 \text{ g}^{-1}$ for **6FT-RCC3**) although the pore diameter was slightly lower than the Van der Waals radius of a N_2 molecule. Furthermore, **6FT-RCC3** retains porosity even after harsh treatment with acids or under basic conditions.

With this knowledge over the control of the inside cavity, the groups of Cooper and Hirscher were able to selectively functionalize the amine group-containing cavity of **RCC3** with aldehydes of different chain lengths to generate sixfold aminal-containing POCs and were also able to mask one position by acetone-tying before reacting the ten remaining amine groups to form five additional aminals.^[14b] Ultimately, they discovered that a sixfold acetaldehyde-tied cage **6ET-RCC3** produced an internal cavity with a diameter of only $1.95 \text{ \AA}$, which is barely enough to fit a H_2 molecule. This was utilized to selectively separate H_2 from D_2 with an initial selectivity of $\text{D}_2/\text{H}_2 \approx 4$. By combining the high capacity of **CC3** with the selectivity of **6ET-RCC3** in a cocrystal, the groups could increase the selectivity D_2/H_2 to ≈ 8 . The resulting porous material is therefore one of the best options to separate these two very similar gases by using a size exclusion approach. The ground-breaking crystalline system could only be prepared successfully, because the functionalization of the inner pore had almost no influence on the intermolecular interactions responsible for the crystal packing. The crystal lattice remained largely unaltered, allowing

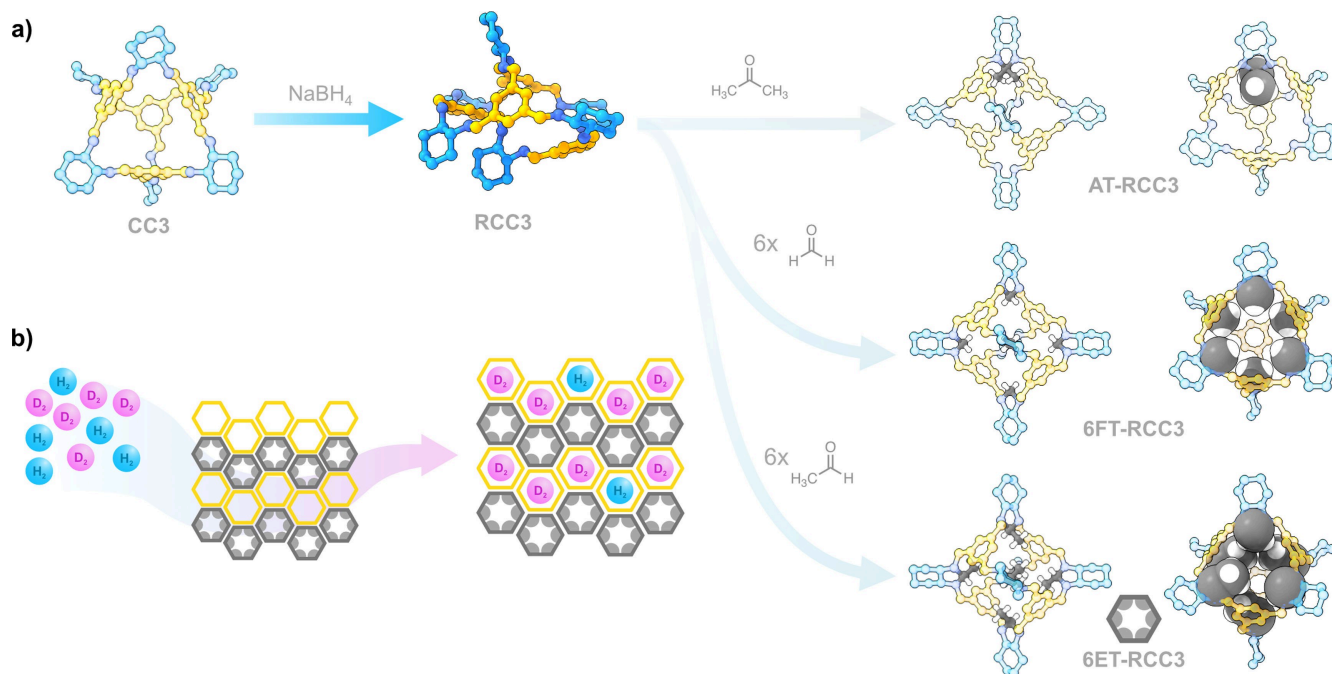


Figure 4. The parent Tri^4Di^6 POC **CC3** by the group of Cooper forms after $[4 + 6]$ imine condensation, when reacting six equivalents of *R*-1 with four equivalents of **3**. a) **CC3** was reduced to the corresponding dodecaamine cage, **RCC3**, by treatment with NaBH_4 in almost quantitative yield.^[14c] The loss of porosity for **RCC3** was ascribed to a collapse of the flexible cage cavity upon desolvation, rather than to it being amorphous, because multiple attempts to desolvate **RCC3** by slow drying, solvent exchange, or supercritical fluid drying all resulted in amorphous solids with desolvated **RCC3** not exhibiting any porosity to either N_2 or H_2 at 77 K , or to CO_2 at 298 K .^[14c] **RCC3** reacted with acetone during recrystallisation to **AT-RCC3** selectively (top), was reacted with paraformaldehyde at 70°C in methanol in 70% yield, tying all six diamine vertices (middle), but also reacted with acetaldehyde to give fully tied **6ET-RCC3** in 83% yield; b) cocrystals combining the high capacity of **CC3** with the selectivity of **6ET-RCC3**, allowing for isotopic sieving of deuterium from hydrogen.^[14b] Hydrogen atoms and solvent molecules are omitted for clarity, space-filling view including hydrogen atoms for *endo*-groups.

for precise control over size exclusion inside the interconnected 3D pores. All gas sorption data is collected in Table 1.

In addition to changing the pore size, inner cavity functionalization also changed the electronic nature of the pore in the following example. When Cooper *et al.* revisited their **6FT-RCC3** cage and investigated the SO₂ binding ability, the groups could show that even though the inner surface area was slightly smaller compared to non-functionalized **CC3** (2.8 mmol g⁻¹), the adsorption of SO₂ increased by a factor of five (13.8 mmol g⁻¹).^[14a] The researchers identified the binding sites to be the nitrogen atoms within the cavity. Since **CC3** is composed of rather electron-deficient imine groups, the ability to bind SO₂ was limited. **RCC3** as well as **6FT-RCC3** contain amine groups in which much more electron density is located at the nitrogen lone pairs. Comparing the amines in **RCC3** with the ones in **6FT-RCC3**, it becomes clear that the tertiary amines of **6FT-RCC3** induced a much higher affinity to SO₂ even though

the pore diameter was much smaller. **6FT-RCC3** also exhibits complete reversibility of SO₂ adsorption over at least 50 cycles.

Mastalerz and co-workers reported a series of transformations of their Tri⁴Di⁶ cage **SC1** that were also inherently stabilized because of the salicyldialdehyde **6**, by so called “imine clip” formation.^[16f] **SC1** is highly porous, and the inner cavity can be modified, as described for the alkylation of exposed hydroxy groups. Tri⁴Di⁶ POC **SC1** was also shown to undergo successful *exo*-functionalization (Figure 5).

Variation of the periphery of the Tri⁴Di⁶ cage led to *S*_{BET} values of around 700 m²g⁻¹ for all amorphous cage compounds, suggesting negligible influence of the bulkiness of the peripheral groups. In their crystalline states, *exo*-groups largely influenced the accessible surface area, allowing for control of the extrinsic pores by crystal engineering.^[16e] To increase the chemical stability of imine cages, reduction to amines was reported but is often accompanied by a higher degree of

Table 1. Overview of gas sorption data available for the CC-series of Tri⁴Di⁶-cages.

	<i>S</i> _{BET} [m ² g ⁻¹]	N ₂ uptake ^[a] [mmol g ⁻¹]	H ₂ uptake ^[a] [mmol g ⁻¹]	CO ₂ uptake ^[b] [mmol g ⁻¹]	CH ₄ uptake ^[b] [mmol g ⁻¹]
CC3 ^[12e]	624	8.2	5.0	2.5	1.5
CC15 ^[12d]	2.7	0.3	2.9	1.3	–
CC3-S/CC15-R ^{[c][12d]}	13.1	0.4	3.4	1.8	–
CC3-RS ^[12a]	598	9.6	5.9	2.0	1.9
CC19-RS ^[12a]	500	8.1	5.7	2.4	1.3
RCC3 ^[14c]	no porosity	no porosity	no porosity	no porosity	no porosity
AT-RCC3 ^[14c]	67	1.1	1.3	1.8 ^[d]	–
6FT-RCC3 ^[14c]	377	11.2	4.3	1.4 ^[d]	–
6ET-RCC3 ^[14b]	–	1.4	H ₂ : 3.8 ^[e] D ₂ : 0.4 ^[e] H ₂ : 2.2 ^[e] D ₂ : 4.7 ^[e]	0.5 ^[d]	–
6ET-RCC3-R/CC3-S ^{[c][14b]}	–	4.0		0.9 ^[d]	–

[a] Measured at 77 K and 1 bar. [b] Measured at 273 K and 1 bar. [c] Cocrystal. [d] Measured at 298 K and 1 bar. [e] Measured at 30 K and 1 bar.

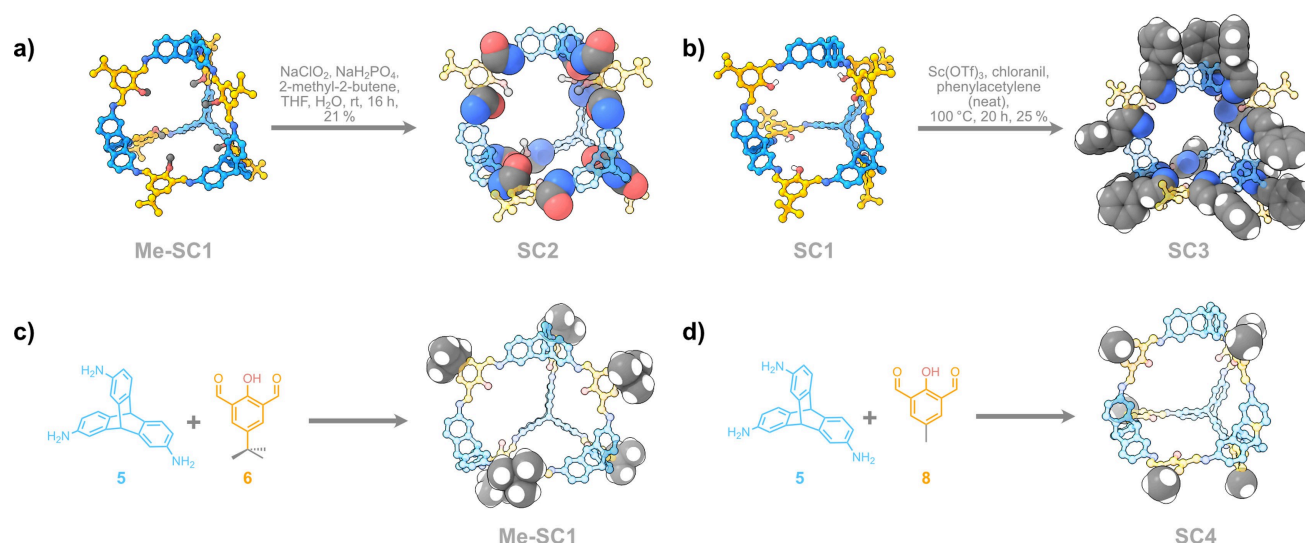


Figure 5. The parent salicylbisimine Tri⁴Di⁶ POC **SC1** by the group of Mastalerz underwent various reactions for *exo*-functionalization. a) After post-synthetic methylation, cage **Me-SC1** underwent Pinnick oxidation, for transformation into the chemically robust amide cage **SC2**.^[16c] b) reaction of the parent POC **SC1** by twelvefold Povarov reaction utilizing phenylacetylene gave quinoline POC **SC3** in one step in 25% yield.^[16b] c) and d) a range of salicyldialdehydes **6** & **8**, bearing various alkyl substituents can be used to generate *exo*-functionalized salicylbisimine POCs, that allowed for successful control of the extrinsic pores in crystalline states.^[16e] Hydrogen atoms and solvent molecules are omitted for clarity, space-filling view including hydrogen atoms for *exo*-groups.

conformational freedom, leading to loss of porosity. Twelvefold Pinnick oxidation was shown to again lead to highly porous and chemically robust amide cages that are also stable over a large pH range, even allowing further post modification by means of direct bromination.^[16c] All sorption data for this class of compounds is presented in Table 2.

In addition to this, the imine cage **SC1** can also be transformed by twelvefold Povarov reaction utilizing phenylacetylene to yield quinoline POC **SC3** in one step. The as-synthesized material was amorphous but porous. This allowed for gas adsorption to be measured again after thorough activation (residual solvent removal), whilst also being acidochromic.^[16b] Independent of the nanocavity within the POCs, three congeners of rather insoluble larger imine cages have been synthesized by attaching solubilizing sidechains to the triptycene triamine precursor recently.^[16a]

From 2018 on, a series of shape-persistent imine cages with a truncated tetrahedral geometry were reported. This Tri⁴Tri⁴ topology is formed from [4 + 4] condensation reactions employing different 1,3,5-triaminomethylbenzene derivatives in combination with 1,3,5-triformylbenzenes by the groups of Mastalerz and Schmidt (Figure 6).^[17]

Substituents on the building blocks were important not only in terms of cavity control, but also in terms of window size and synthesis yield. Often employed, triethyl-substituted triamine **9** shows a rotational barrier of +227.7 kcal mol⁻¹ around the C_{sp2}–C_{sp3} bond of the methylamine substituent, providing rigid preorganization for facile cage formation to enable the formation of **TC1** (47%), **TC2** (37%) and **TC3** (27%).^[17b,d] Single-crystal X-ray diffraction (SXRD) and specific surface area analysis

by N₂ sorption at 77 K showed that in the solid-state the cage windows of **TC1** are blocked by the additional alkyl substituents. For **TC3** on the other hand, the gas sorption behavior changed significantly when activated with liquid ethane at 157 K for ten minutes, before drying. The specific surface area became much larger with 443 m² g⁻¹ and **TC3** adsorbs 4.7 mmol g⁻¹ H₂ (at 77 K, 1 bar), 3.1 mmol g⁻¹ CO₂ and 1.4 mmol g⁻¹ CH₄ (both at 273 K, 1 bar) (Table 3).^[17d] Additionally, the series of Tri⁴Tri⁴ truncated tetrahedral cages was thoroughly investigated with respect to the window apertures as hosts for various tetra-*n*-alkylammonium salts of different sizes.^[17b] Schmidt and co-workers synthesized a Janus-like ("two-faced") POC **TC4**, employing 1,3,5-tris(aminomethyl)-2,4,6-trifluorobenzene **12** in combination with 1,3,5-triformylbenzene **3**.^[17a] Although the preorganization is significantly lowered to only 7 kcal mol⁻¹, **TC4** can be obtained as crystalline material in good yields of 67% without addition of acid from DCM/MeOH solution. In accordance with the observations of Mastalerz and co-workers,^[17b,d] all Tri⁴Tri⁴ cages discussed herein might indeed be the kinetic products. Opening the windows and modifying the nanocavity of **TC4** by the addition of electron-withdrawing fluorine atoms, the POC showed indeed improved gas sorption behavior after drying the crystal filtered off of the reaction mixture. The specific surface area for the crystalline material of 536 m² g⁻¹ (derived from the BET isotherm) agreed very well with a microporous type I model. **TC4** showed a higher uptake of up to 4.2 mmol g⁻¹ CO₂ (at 273 K, 1 bar) and 7.5 mmol g⁻¹ H₂ (at 77 K, 1 bar) compared to similar sized **CC2** (3.0 mmol g⁻¹ CO₂, 6.0 mmol g⁻¹ H₂) and **CC3** (2.5 mmol g⁻¹ CO₂, 5.0 mmol g⁻¹ H₂).^[13d] An explanation for the higher gas uptake is likely found

Table 2. Overview of gas sorption data available for the SC-series of Tri⁴Di⁶-cages.

	S _A ^{BET} [m ² g ⁻¹]	N ₂ uptake ^[a] [mmol g ⁻¹]	H ₂ uptake ^[a] [mmol g ⁻¹]	CO ₂ uptake ^[b] [mmol g ⁻¹]	CH ₄ uptake ^[b] [mmol g ⁻¹]
SC1 ^[16g]	1377	19.7	5.6	2.1	0.6
	2071 ^[c]	27.2 ^[c]			
SC4 ^[16e]	729	13.2	–	–	–
	1291 ^[c]	15.4 ^[c]			
Me-SC1 ^[14d]	824	15.4	–	–	–
	1700 ^[c]	24.1 ^[c]			
n-Pr-SC1 ^[14d]	494	5.9	–	–	–
allyl-SC1 ^[14d]	333	4.8	–	–	–
Bz-SC1 ^[14d]	119	1.3	–	–	–
SC1_L ^[16g]	< 1	–	–	–	–
SC2 ^[16c]	275	8.8	4.0	2.1	0.7
OH-SC2 ^[16c]	398	14.5	3.0	2.0	0.5
SC3 ^[16b]	698	19.3	5.2	2.7	1.2

All sorption data from the isolated compounds, if not described otherwise. Amine cage **SC1_L** can be obtained by sodium borohydride reduction of **SC1**.^[16g]

[a] Measured at 77 K and 1 bar. [b] Measured at 273 K and 1 bar. [c] Recrystallized sample.

Table 3. Overview of gas sorption data available for the TC-series of Tri⁴Tri⁴-cages.

	S _A ^{BET} [m ² g ⁻¹]	N ₂ uptake ^[a] [mmol g ⁻¹]	H ₂ uptake ^[a] [mmol g ⁻¹]	CO ₂ uptake ^[b] [mmol g ⁻¹]	CH ₄ uptake ^[b] [mmol g ⁻¹]
TC1 ^[17d]	71 ^[c]	2.6 ^[c]	1.7 ^[c]	1.9 ^[c]	0.8 ^[c]
TC3 ^[17d]	443 ^[c]	6.1 ^[c]	4.7 ^[c]	3.1 ^[c]	1.4 ^[c]
TC4 ^[17a]	537	7.6	7.5	4.2	–

No sorption data has been reported for **TC2**.^[17b] [a] Measured at 77 K and 1 bar. [b] Measured at 273 K and 1 bar. [c] Activated by washing with liquid ethane at 157 K for 10 min and dried at 40 °C in vacuum prior to measurement.

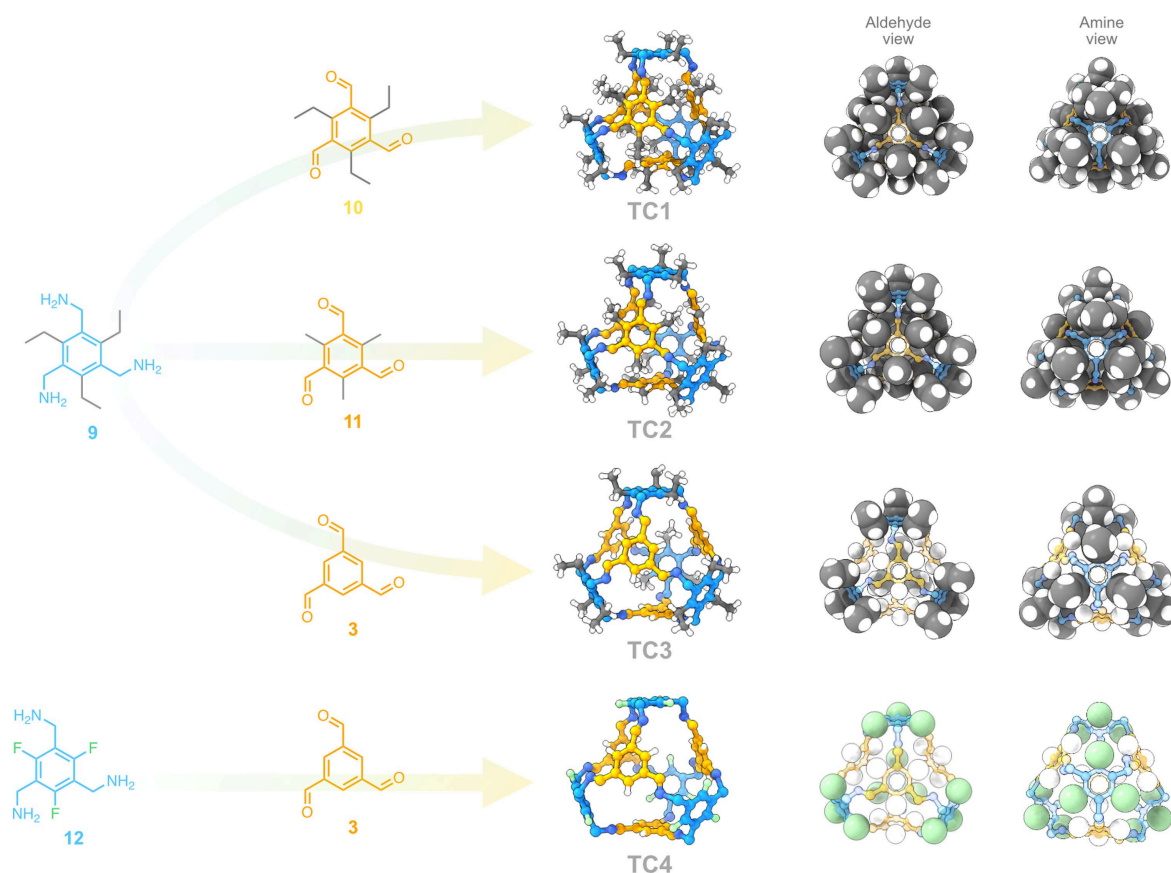


Figure 6. Truncated tetrahedral cages following Tri^4Tri^4 topology were formed from [4 + 4] condensation reactions employing functionalized 1,3,5-triaminomethylbenzene derivatives in combination with different 1,3,5-triformylbenzenes.^[17] A series of POCs bearing alkyl substituents were shown to assemble in acetonitrile giving **TC1** (Et–Et, 47%), **TC2** (Et–Me, 37%) and **TC3** (Et–H, 27%). The group of Schmidt synthesized 1,3,5-tris(aminomethyl)-2,4,6-trifluorobenzene **12**, to generate Janus-like POC **TC4** (F–H) in 67% yield. Hydrogen atoms and solvent molecules are omitted for clarity, space-filling view including hydrogen atoms for showcasing the influence of adjacent groups on the window size.

in the higher hydrophobicity of the nanospaces due to the fluorinated aromatic panels of Janus-like **TC4**, that are also less sterically demanding (Figure 6 bottom).

Only electron-deficient benzonitrile, hexafluorobenzene, and *para*-iodo-nitrobenzene were able to form strong enough host-guest complexes that survived the electrospray ionization process without complete dissociation, whereas no complexation-induced chemical shift changes were observed in NMR studies. In addition, eight hybrid Tri_4Di_6 imine cages with tunable fluorination and one fully fluorinated Tri_4Di_6 imine cage were studied. Despite the vast differences in reactivity between fluorinated and non-fluorinated building blocks, it was possible to gain control over the outcome of the self-assembly process by carefully controlling the feed ratio.

After recrystallization, these cages with unlimited miscibility in the solid-state were obtained as highly crystalline samples that even retained the crystal lattice, forming alloys. They remained essentially nonporous although huge voids are present in the structure, because they are not interconnected, OD pores.^[18]

A series of three rylene-type cages were investigated by the group of Šolomek, also extending the repertoire of POCs towards photochemistry (Figure 7).^[19] The incorporated rylene

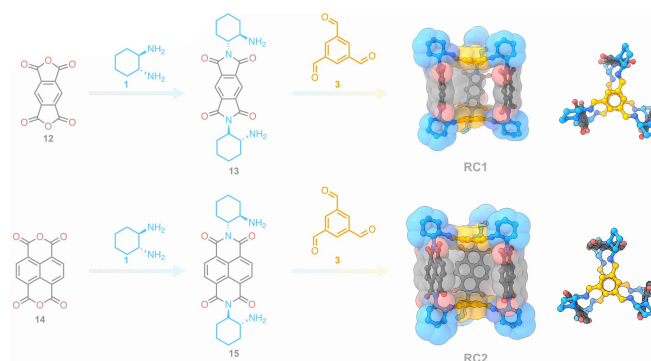


Figure 7. The synthesis of chiral Tri^2Di^3 POCs incorporating rylene diimides and their molecular structures determined from SXRD. Hydrogen atoms and solvent molecules are omitted for clarity.^[19]

diimides are not only interesting because of their strong adsorption and emission but they are also potent electron acceptors with favorable redox potentials.

All chiral Tri^2Di^3 POCs **RC1–RC3** were thoroughly investigated by SXRD, exhibiting edge-to-face packing with a herringbone-like structure.^[20] Cage **RC2** showed permanent

porosity and a type I adsorption isotherm for ($SA_{\text{BET}} = 522 \text{ m}^2 \text{ g}^{-1}$) upon activation by dynamic vacuum or with supercritical CO_2 . The small difference in building block size led to cage **RC2** being shape-persistent, whereas **RC1** showed a gate opening phenomenon, usually observed for soft porous crystals and flexible porous cages, where fast uptake of N_2 only occurred after a threshold pressure was reached. Cages **RC1** and **RC2** also both showed higher affinity towards CO_2 , with capacities of 2.0 and 2.9 mmol g^{-1} , respectively. Moreover, they even showed CO_2/N_2 selectivity and CH_4 uptake.^[19b] These results also suggest the benefits of precise pore control and electron-deficient nanocavities, complementing ongoing POC research.

The bulk of work so far described imine condensation reactions for POC self-assembly but boronate ester formation, among others, is also a powerful tool to generate porous organic materials.^[9,21b] The group of Beuerle contributed several giant molecular cubes, assembled from catechol-functionalized tribenzotriquinacenes and 1,4-phenylene diboronic acids, which were synthesized in one-pot procedures by crosslinking 20 individual components through a dynamic covalent approach.^[4a,6a,21c] Investigations on the self-sorting of ternary mixtures containing two competitive boronic acids were reported later.^[21b] The group introduced an additional supramolecular recognition side into the diboronic acid. The benzoic acid group incorporated was shown to lead to a perfectly preorganized circular hydrogen-bonding array within the nanocavity of a trigonal-bipyramidal Tri^2Di^3 cage, whose formation was energetically overcompensated by the additional strain energy caused by the larger mismatch in bite angles for the smaller assembly.^[21a] The group of Mastalerz also reported a small series of shape-persistent Tri^4Di^6 boronic ester cages containing bromide and fluoride substituents in an attempt to influence the intrinsic and extrinsic pore (Figure 8).^[21d]

The brominated Tri^4Di^6 boronic ester cage **BC1** became porous after activation ($SA_{\text{BET}} = 511 \text{ m}^2 \text{ g}^{-1}$) and showed unique

gas-sorption selectivities of saturated hydrocarbons over unsaturated hydrocarbons. Henry selectivities revealed the selective adsorption of ethane over ethylene and acetylene with $S_{\text{H}} = 1.9$ ($\text{C}_2\text{H}_6/\text{C}_2\text{H}_4$) and $S_{\text{H}} = 3.1$ ($\text{C}_2\text{H}_6/\text{C}_2\text{H}_2$) and consequently a selectivity of ethylene over acetylene of $S_{\text{H}} = 1.6$ ($\text{C}_2\text{H}_4/\text{C}_2\text{H}_2$).^[21d]

3. Trianglimines

Trianglimines form a class of often non-racemic macrocycles which, similar to most POCs, are based on the formation of imine bonds. The name-giving, triangular geometry is the result of a sixfold condensation reaction between a ditopic chiral amine (mostly *trans*-1,2-diaminocyclohexane) and ditopic aldehydes, giving Di^3Di^3 macrocycles. The first members of this class were reported over two decades ago by Gawronski *et al.*, who could obtain the terephthalaldehyde (**20** & **21**) based trianglimines **T1** and **T2** in yields around 90% with high purity, by utilizing the limited conformational freedom of the imine bonds and the structural preorganization induced by diaminocyclohexane **1** (Figure 9).^[22b] Following these initial results, a variety of articles were published in close succession, mainly focusing on the synthetical aspects,^[22–23] such as diversification by formaldehyde tying,^[22a] the inherent chirality or host-guest interactions in solution,^[23] obtained crystal structures mainly being a supplement. Although even in the earliest works either inclusion complexes or pores, when omitting the solvent from the structure, could be observed in the solid-state,^[22c,24] it took another decade until groups started focusing on the solid-state adsorption properties.^[24–27] Other than POCs with their inherent 3-dimensional cavity, in the 2-dimensional structure of macrocycles only a planar void (hole) is present. If multiple units are stacked in a layered fashion, then tubular 1D pores (channels) are formed.

Homogenous channels where the pore size is equal throughout the entire structure, can potentially be used for

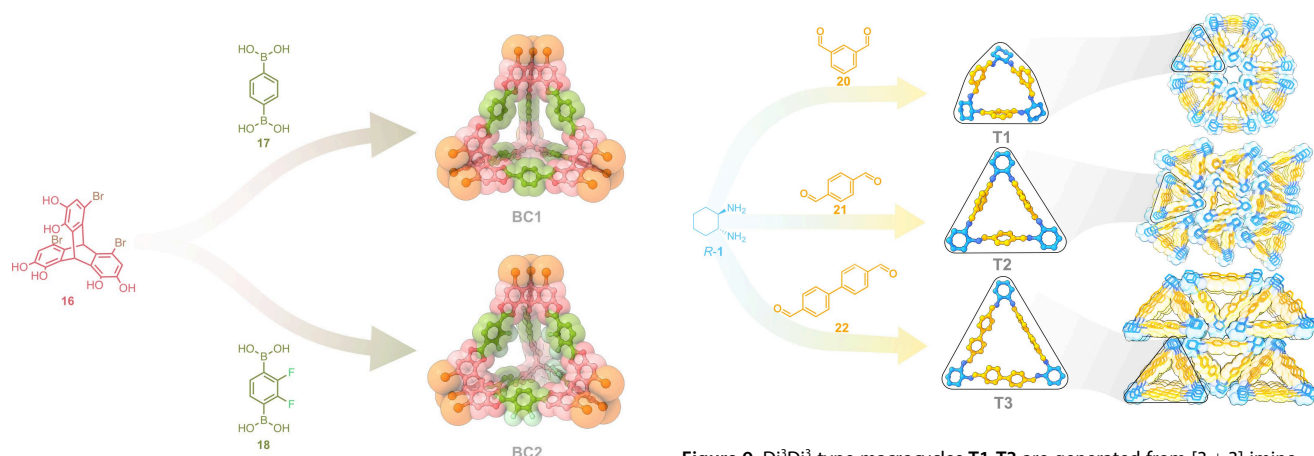


Figure 9. Di^3Di^3 -type macrocycles **T1**–**T3** are generated from [3 + 3] imine condensation employing a ditopic chiral amine **1** and various ditopic aldehydes **20**–**22**. Molecular structures showing the planar voids present (encircled) and solid-state superstructures (right) determined from SXRD show the formation of tubular 1D pores, when multiple units are stacked in a layered fashion. Hydrogen atoms and solvent molecules omitted for clarity.^[25–26]

separation by host-guest interactions in the solid-state. As described for POCs, it can also be challenging to achieve a permanently porous structure, as the macrocycles need to maintain the structure upon activation (residual solvent removal). In 2016 Janiak *et al.* were able to show that the crystal structure of the non-porous formaldehyde tied TA-based trianglamine **FT-T2** could be transformed into a porous structure.^[24] By using primary alcohols belonging to the homologous series from ethanol to *n*-octanol, they observed that for short-chain alcohols, inclusion complexes without interconnected pores were formed, while long-chain alcohols acted as strings, threading the macrocycles while forming long interconnected 1D channels. After activation, the desolvated crystals maintained the porous structure and were able to adsorb EtOH in a 1:1 molar ratio.

To understand the type of supramolecular interactions that govern the recognition of benzene derivatives, Khashab *et al.* investigated the influence of guests on intermolecular interactions using toluene or mesitylene as guests for the 4,4'-diformylbiphenyl (**22**) based macrocycle **T3**.^[25c] Crystallizing **T3** in presence of these guests, lead to the observation of different kinetically favored, metastable superstructures. In both cases initially, fibrous organogels were formed upon dissolving the macrocycle, which redissolved over 14 days at room temperature forming either a capsule-like (mesitylene) or helical (toluene) superstructure in solution, of which the thermodynamically stable needle-shaped (mesitylene) or block-shaped (toluene) crystals were obtained. SXRD revealed that the macrocycle crystallized with the respective guest binding in the cavity generated by the macrocycles. The overall packing was deemed to originate from C–H... π interactions between the methyl groups of the guests and the aromatic panels of **T3**. Based on these observations, the group shortly after reported the efficient separation of styrene (ST) from ethylbenzene (EB) in both liquid and vapor phase using **T3** (Figure 10).^[25a] Crystals grown from a ST/EB mixture (1:1 v/v) formed the inclusion complex **ST@T3**, revealed by PXRD, showing a 99.9% selectivity for ST-uptake when submitting the crystals to GC analysis. In solid-vapor experiments, the crystalline macrocycle readily adsorbed EB and ST individually, while time dependent studies clearly showed that the uptake of ST (8 h) was significantly faster than for EB (24 h). Additional vapor diffusion experiments using ST/EB (1:1 v/v) mixtures showed 88% selectivity towards ST, even increasing to 99%, when using 4:1 (v/v) ST/EB mixture.

The selectivity was reasoned to stem from the non-covalent interactions between the host and guest, as PXRD studies of solid-vapor experiments revealed that **T3** undergoes a structural transformation towards the inclusion complexes **ST@T3** or **EB@T3** respectively. The planar ST underwent only slight structural changes, while the host-guest interactions were maximized in comparison to the non-planar EB.

Similar observations were made by Cooper and co-workers, who recently studied the TA based macrocycle **T2** in respect to its adsorption and separation properties (Figure 10 bottom).^[26a] When crystallizing **T2** from ethyl acetate (EA), the group observed a 1:1 molar inclusion complex of **EA@T2**, in which EA is strongly bound inside the generated 1D pores, indicated also

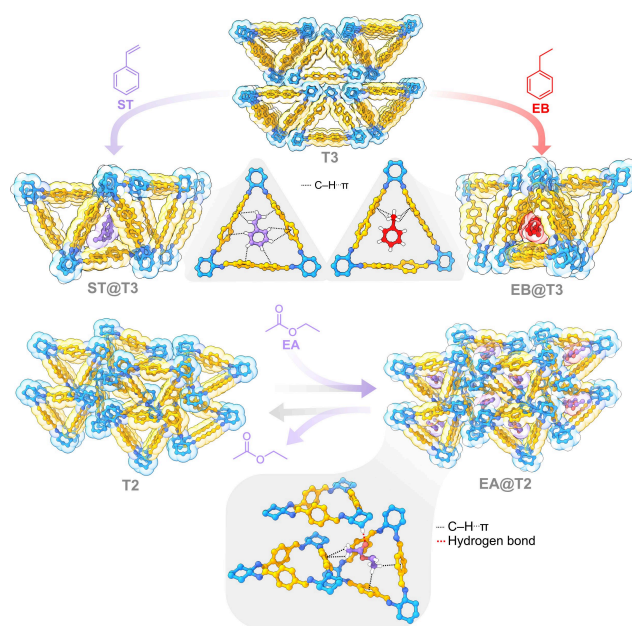


Figure 10. Crystalline macrocycle stacks of **T3** readily adsorbed EB and ST individually in the intrinsic pore, facilitated by C–H... π interactions between the methyl groups of the guests.^[25a] Ethyl acetate (EA) is strongly bound inside the generated porous channels of **T2**, forming a 1:1 inclusion complex **EA@T2**. The necessity of the activation of **EA@T2** at elevated temperatures under high vacuum for elongated time is proof of the very firm binding of the EA molecules inside the small pores.^[26a] Host-guest structures determined from SXRD with hydrogen atoms of the host macrocycles omitted for clarity.

by the comparatively harsh activation conditions needed to remove the guest, ultimately yielding the guest-free structure. SXRD analysis revealed that i) the high affinity is attributed to the formation of hydrogen bonds and C–H... π interactions, while EA also fits perfectly into the 1D pore and ii) the guest-free structure was observed to be nearly isostructural to **EA@T2**, maintaining the complementary voids for EA. Solid-vapor experiments with EA/EtOH (1:1 v/v) showed the anticipated higher and faster uptake of EA (0.8 mol mol⁻¹ within 120 min) in relation to EtOH (0.2 mol mol⁻¹ within 120 min). The system equilibrated over 14 h fully, replacing the adsorbed EtOH with EA, indicating the exceptional selectivity of the system. Break-through experiments using EA/EtOH vapor (1:1 v/v) underlined the previously observed selectivity and affinity, as EtOH breaks through over the period of 50–300 minutes, while EA did not break through until 300 minutes, reaching feed concentration after 500 minutes.

Those examples highlight the great potential of separating small molecules of industrial and chemical interest through shape specific host-guest interactions. On the downside, the systems simultaneously showed relatively slow adsorption kinetics, supposedly due to the triggered structural change upon guest adsorption. Hence, Cooper *et al.* recently revisited the *iso*-TA based macrocycle **T1** regarding the porosity of the system, speculating that the adsorption kinetics can be enhanced.^[26b] Gas adsorption experiments with *R*-**T1** showed that activation of the crystalline sample resulted in a basically

non-porous polycrystalline material ($SA_{\text{BET}} \approx 4 \text{ m}^2 \text{ g}^{-1}$), due to collapsing of the pores upon solvent removal. Similar to **CC3**,^[9] co-crystallization using equimolar amounts of the compound with opposed chirality (**S-T1**) lead to a window to window packing arrangement for each heterochiral pair (Figure 11).

The pores generated in the cocrystal **rac-T1**, were stabilized by π - π stacking and C-H... π interactions between each racemic substructure, also maintaining the overall pore structure after activation. The resulting material was shown to be porous with an apparent $SA_{\text{BET}} = 355 \text{ m}^2 \text{ g}^{-1}$. Measured gas uptakes were significantly higher for **rac-T1** in comparison to homochiral **R-T1** for H_2 (2.6 mmol g^{-1} vs. 0.1 mmol g^{-1} at 1 bar and 77 K) as well as for CO_2 (3.8 mmol g^{-1} vs. 2.2 mmol g^{-1} at 1 bar and 195 K). Further showing a faster adsorption kinetic (3 h until saturation) in time dependent solid-vapor experiments with a *para*-(**pX**) and *meta*-xylene (**mX**), reaching maximum uptake after 10 h (0.8 mol pX per mol **rac-T1**) presumably attributed to the higher porosity achieved by the heterochiral pairing. While the observed high selectivity coefficient of $K_{\text{pX,mX}} = 15.7$ was reasoned to originate from the minimal structural difference between the crystal structure of inclusion complex **pX@rac-T1** and the guest free structure of **mX@rac-T1**, where a different packing mode was observed when the guests were added to crystallization mixture.

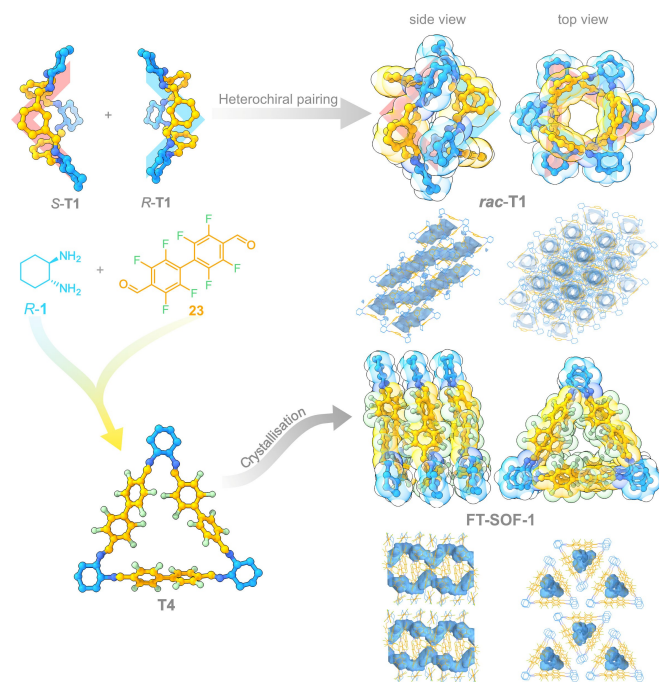


Figure 11. Although **T1** molecules stacked in an eclipsed fashion along the crystallographic *c*-axis, forming pillars of **T1**, the pore structure collapsed after drying. Cococrystallizing **S-T1** with **R-T1** lead to heterochiral window-to-window packing, with a pore structure of sufficient stability to enable residual solvent removal.^[26b] Solid-state packing with accessible voids depicted in dark blue. A highly fluorinated triamine macrocycle **T4** was synthesized and exhibits a chiral helical pore in the solid-state with again accessible voids depicted in dark blue, stable also upon desolvation.^[27] Structures determined from SXRD with hydrogen atoms and solvent molecules omitted for clarity.

In contrast to the previous examples, Schmidt *et al.* reported the first highly fluorinated triamine based SOF by employing aldehyde **23**, altering the steric and electronic effects of the used building block.^[27] The fluorinated triamine SOF **FT-SOF-1** was formed by simple crystallization of triamine **T4** from a DCM/MeCN solution as long, needle-shaped crystals. Interlayer C-F... π and C-H...F interactions,^[20] as well as the twisted fluorinated biphenyl panels, caused by the higher steric demands of the fluorine substituents enabled the formation of a unique helical 1D pore (Figure 11 bottom). **FT-SOF-1** retained its porosity even after activation, as the moderate $SA_{\text{BET}} = 88 \text{ m}^2 \text{ g}^{-1}$ indicates, even though the inner pore diameter is reduced due to the helical arrangement and larger fluorine atoms. The unique shape, size and electronic properties of the channels make it a promising material in regard to CO_2/CH_4 separation.

In addition to the rigid triamines, Khashab *et al.* showed the exceptional CO_2/CH_4 gas separation properties when employing the reduced congeners, dubbed triaminamines, for separation studies. These hexamines were simultaneously introduced with the hexamines in works by Gawronski *et al.*^[22b] They could be obtained in nearly quantitative yields by reduction with sodium borohydride. Although chemically more stable, they suffer from the flexibility of the amines. To counteract this flexibility, they can again be transformed into the more rigid, tied triaminamines,^[22a,24] following the discussed method of cage tying. Khashab *et al.* managed to obtain a permanently porous triamine-based SOF (**T-SOF-1**) utilizing the reduced congener of **T2**, the hexamine **T2_L** (Figure 12).^[25d] The flexible amine was crystallized after slow diffusion of acetone vapors into an aqueous HCl solution of the macrocycle. SXRD revealed that due to the strong directing effects of chloride counterions, the macrocycles were arranged in a

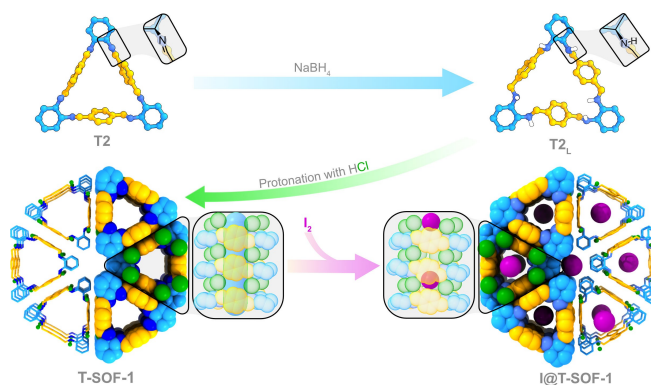


Figure 12. Reduction of triamine **T2** yields triamine **T2_L**. SXRD structure of **T-SOF-1**, showing the hexagonal arrangement of triaminamines **T2_L** schematically after protonation and their tubular packing forming triangular channels running along the *c*-axis, with chloride counterions as directing-agents. **T-SOF-1** crystals soaked in a solution of iodine in acetonitrile for 3 h, yielding **I@T-SOF-1**. The quality of the collected data was not sufficient to determine the nature of the iodo-species entrapped within the channels, however, Raman studies revealed the presence of polyiodide (I_3^- and I_5^-) rather than iodine.^[25d] All molecular structures determined from SXRD with hydrogen atoms and solvent molecules omitted for clarity.

porous hexagonal shaped structure generating two tubular channels, one intrinsic (~ 6.3 Å) and one extrinsic (4.5 Å).

After solvent exchange to MeCN and activation in vacuum, the **T-SOF-1** with a S_{BET} of $170 \text{ m}^2 \text{ g}^{-1}$ was obtained. **T-SOF-1** showed a high affinity towards CO_2 (1.0 mmol g^{-1}), while also showing a moderate adsorption of CH_4 (0.5 mmol g^{-1}) at 298 K and 1 bar. **T-SOF-1** encapsulated iodide molecules, resulting in the crystalline inclusion complex **I@T-SOF-1** in which the iodide is located in both channels, effectively reducing the pore diameter. Surprisingly, following gas adsorption studies showed only a marginal decrease in CO_2 uptake (0.9 mmol g^{-1}), while simultaneously displaying negligible CH_4 uptake at ambient conditions, demonstrating an interesting methodology to fine-tune the size, properties and therefore selectivity of formerly rather unselective pores.

Trianglaminates open up the possibility to post-synthetic modifications, for example tying neighboring amines with either ketones^[22a] or transformation into the related thiourea motifs.^[28] Additionally, *N*-alkylation or acylation of the hexa- amines using the respective halides can be used to introduce new functionalities.^[22a,23b] Khashab *et al.* used terephthaloyl chloride (TPC) to prepare highly crosslinked membranes by interfacial polymerization with **T₂** as monomer. The ultra-thin, selective layer (< 10 nm) with a small molecular weight cut-off (around 450 g mol^{-1}) was able to separate chiral solutes (amino acids) by chirality due to the specific pore size and chiral nature exhibited by the trianglamine moiety.^[25b] During the production of this minireview, studies on *m*-xylene separation,^[25e] and haloalkane isomer separation^[25f] appeared. Trianglaminates and their congeners show great potential and synthetic diversity by using a rather simplistic motif, further demonstrating that going a step back to the simpler, discrete two-dimensional structures can be a very promising approach.^[29]

4. Cyclobenzoin

Although this review focuses predominantly on systems generated by dynamic covalent chemistry with aldehydes mostly perceived as imine precursors, in 2015 Miljanić *et al.* discovered a new class of macrocyclic compounds dubbed cyclobenzoin, which are prepared by benzoin condensation of aromatic dialdehydes.^[30h,i] A cyclic trimer (cyclotribenzoin **B1**) or a cyclic tetramer (cyclotetrazobenzoin **B2**) was formed when using **20** & **21**, respectively.^[30g,h] While the trimer did not exhibit a porous structure, the square-shaped **B2** formed similarly shaped 1D pores due to zigzag hydrogen bonding connections with a different layer of the neighboring column (Figure 13).

This arrangement resulted in an overall porous crystal ($S_{\text{BET}} \approx 50 \text{ m}^2 \text{ g}^{-1}$). Once synthesized, the reactive hydroxyl groups could be directly silylated^[30h] or esterified.^[30c,d] Using the latter method, the S_{BET} could be drastically improved. Esterification with acetic anhydride resulted in a permanently porous material (**Ac-B2**) upon crystallization, exhibiting a type 1 isotherm and a S_{BET} of $572 \text{ m}^2 \text{ g}^{-1}$. Further gas adsorption studies displayed a selective uptake of CO_2 (3.0 mmol g^{-1}) over N_2 (0.1 mmol g^{-1}) and CH_4 (0.4 mmol g^{-1}), with a moderate isosteric heat for CO_2

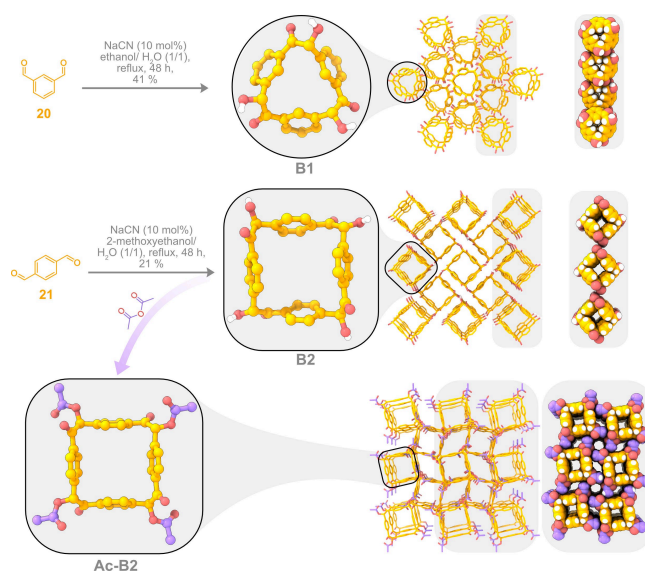


Figure 13. Cyclobenzoin macrocycles prepared by benzoin condensation of aromatic dialdehydes **20** & **21**, their molecular structure and solid-state packing. While the cyclic trimer (cyclotribenzoin **B1**) was not porous (above), the square-shaped **B2** formed similarly shaped 1D pores by zigzag hydrogen bonding connections with a different layer of the neighbouring column (middle). Esterification with acetic anhydride resulted also in a permanently porous material by solid-state assembly of **Ac-B2** (introduced ester moieties depicted in purple). All molecular structures determined from SXRD with hydrogen atoms and solvent molecules omitted for clarity, space-filling views (right) include hydrogen atoms to depict accessible pores.^[30]

adsorption ($Q_{\text{st}} = 33.7 \text{ kJ mol}^{-1}$). Additionally, showing an efficient separation of CO_2/N_2 and CO_2/CH_4 mixtures in column breakthrough and pressure swing adsorptions experiments with over 99.9% and 99.7% selectivity for the gas mixtures, respectively.^[30a] These results make **Ac-B2** a promising candidate for CO_2 capture, flue gas separation and natural gas upgrading. In addition, the benzoin motif present in cyclobenzoin could either be reduced (NaBH_4)^[30h] or oxidized (HNO_3) to the corresponding hydrobenzoin^[30e] or benzils.^[30f] Hence further modifications by imine condensation with *ortho*-diamines allow property fine tuning or crosslinking of the macrocycles to form POPs.^[30b] Generally, the low-cost, easy synthesis and the diversity exhibited by cyclobenzoin and their derivatives make them promising materials of industrial relevance.

5. Conclusion

The research showcased in this review demonstrates the impressive tuneability and performance of ultra-small to medium-sized POCs and macrocycles for selective guest uptake or separation. Several groups demonstrated successfully that general post functionalization strategies are a powerful tool to overcome challenges, such as the ability to separate different feedstock chemicals. Precisely engineered POCs and the subsequent combination thereof, for example by extending the material space by formation of cocrystals or core-shell materials,

have been shown to be extremely efficient, as highlighted by several examples presented herein. Pushing the boundaries of kinetic quantum sieving, a POC which by itself showed very low gas sorption capacities, was converted into a material with optimal gas separation properties. This was achieved by pairing a cage with a small pore with another POC in a single cocrystal with suitable modifications by multistep organic synthesis, enabling high selectivity in combination with high D₂ uptake. In addition to this, macrocycles are a revitalized class of intrinsically porous organic molecules that can also selectively host guests, making them highly valuable building blocks for SOFs or membranes for complex and efficient separations. Continuous research efforts in this field will hopefully further greatly stimulate enhanced and novel applications. In combination with upscaling material preparation, this will most certainly facilitate the translation of this class of materials into technologies, ideally replacing industrial processes with ones having higher energy efficiency regarding separation and purification of commodity chemicals and gases.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: Adsorption • Fluorine chemistry • Materials • Porous organic cages • Supramolecular chemistry

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